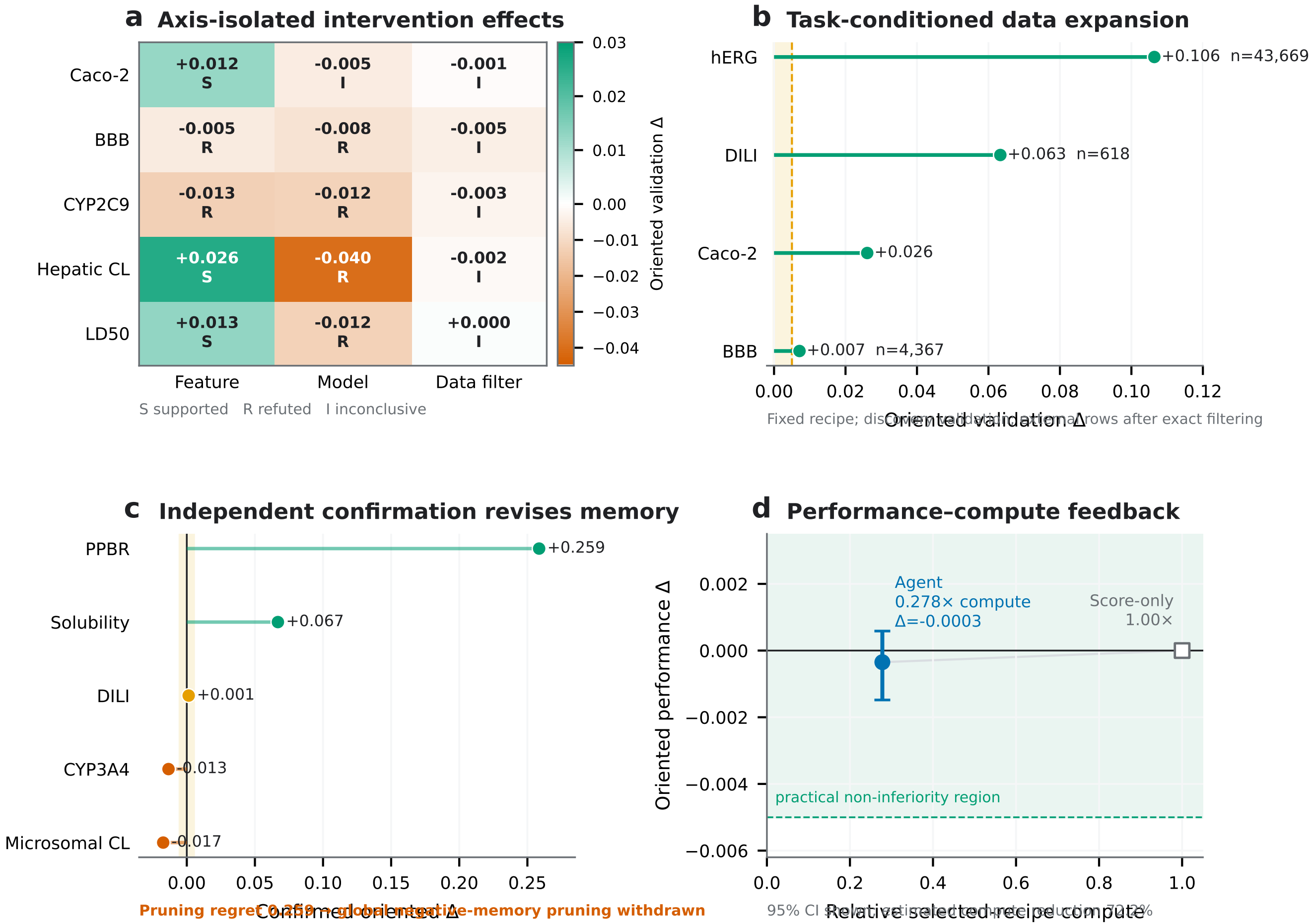
a
Axis-isolated intervention effects
Caco-2
BBB
CYP2C9
Hepatic CL
LD50
+0.012 S
-0.005 I
-0.001 I
-0.005 R
-0.008 R
-0.005 I
-0.013 R
-0.012 R
-0.003 I
+0.026 S
-0.040 R
-0.002 I
+0.013 S
-0.012 R
+0.000 I
Feature
Model
Data filter
S supported R refuted I inconclusive
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
−0.04
Oriented validation Δ
b
Task-conditioned data expansion
hERG
DILI
Caco-2
BBB
+0.106 n=43,669
+0.063 n=618
+0.026
+0.007 n=4,367
0.00
0.02
0.04
0.06
0.08
0.10
0.12
Oriented validation Δ
Fixed recipe; discovery validation external rows after exact filtering
c
Independent confirmation revises memory
PPBR
Solubility
DILI
CYP3A4
Microsomal CL
+0.259
+0.067
+0.001
-0.013
-0.017
0.00
0.05
0.10
0.15
0.20
0.25
Confirmed oriented Δ
Pruning regret 0.259 → global negative-memory pruning withdrawn
d
Performance–compute feedback
Agent
0.278× compute
Δ=-0.0003
Score-only
1.00×
practical non-inferiority region
0.002
0.000
−0.002
−0.004
−0.006
Oriented performance Δ
0.0
0.2
0.4
0.6
0.8
1.0
Relative selected-recipe compute
95% CI shown; estimated compute reduction 72.2%